

Mahmoud Metwalli

Senior Backend Engineer — Node.js / TypeScript

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SUMMARY

Results-driven Senior Backend Engineer specialized in Node.js (NestJS, Express.js) with 4+ years of experience owning and scaling high-traffic consumer applications, and, most recently, building core services for a multi-vendor education marketplace on NestJS, GraphQL, and CQRS. Proven track record of modernizing legacy codebases, reducing critical bugs by 40%, cutting API latency by 45%, and delivering zero-downtime major API version upgrades while maintaining 99.99% uptime. Specialized in REST/GraphQL APIs, event-driven architecture, and secure payment/auth integrations. Also proficient in Python (FastAPI) for backend services.

EXPERIENCE

Senior Backend Engineer • EduMarket

Feb 2026 – Present

- Own core services for a bilingual (AR/EN) multi-vendor education marketplace connecting parents with nurseries, schools, tutors, and transportation providers — a CQRS + hexagonal (DDD) modular monolith across 58 NestJS modules (GraphQL + REST).
- Drove a data-layer and architecture rebuild that lifted throughput 6.8× (4.85 → 32.8 req/s) and cut p95 latency ~6× (60s → 9.9s) under 300 concurrent users, dropping the HTTP failure rate from 79.5% to ~0%.
- Diagnosed bottlenecks with pg_stat_statements and eliminated unindexed sequential scans and N+1 fan-out — top queries fell from 6–15s to sub-millisecond — via targeted read-path indexes, DataLoader batching, and tag-based Redis query caching (99.9% cache-hit rate).
- Restored previously failing enrollment and wallet flows from 0% to 100% success under load, and removed all circular module dependencies (41 → 0), enforced by dependency-cruiser + circular-dep pre-commit/pre-push gates.
- Integrate Kashier and Paymob payment gateways behind cockatiel circuit breakers, alongside wallet and installment/financing modules.
- Safeguard reliability with 3,000+ automated tests across 428 suites (Jest, Testcontainers) and a 23-scenario k6 load-testing suite run against a 1,000,000-user seeded dataset (20→200 RPS spike, 15 RPS/45-min soak), with Sentry/Prometheus/Loki observability.

Mid Senior Backend Engineer • Vconnct

March 2025 – Feb 2026

- Implemented real-time video/audio communication using WebRTC, reliably supporting up to 80 concurrent participants.
- Integrated backend with a dedicated SFU to manage signaling, achieving 25% lower latency and 99.7% connection success.
- Built in-meeting chat module with WebSockets for real-time messaging, typing indicators, and user presence.
- Designed persistent message storage powered by RabbitMQ handling 15k-20k messages/day with <150ms delivery.
- Implemented live transcription pipeline using WebSocket streaming to AI service with 1-2 second latency.

Backend Developer • Ramses International LLC

April 2024 – March 2025

- Designed and built scalable microservices handling peak loads of 2K+ RPS with 99.9% uptime.
- Introduced event-driven architecture using RabbitMQ, reducing synchronous blocking calls by ~70%.
- Built MQTT-based real-time communication layer supporting 5K+ concurrent IoT devices with <500ms latency.
- Engineered co-working space reservation system with distributed locks, handling 1K+ daily bookings.
- Added comprehensive logging and distributed tracing, reducing incident debugging time by 60%.

Backend Developer • TLC Care Services

March 2022 – March 2024

- Modernized legacy codebase for a 10K MAU app, reducing critical bugs by 40% and API latency by 45%.
- Designed and rolled out two major versioned REST APIs (v2 & v3) with zero-downtime deployments.
- Built 40+ high-traffic REST APIs handling 5M+ daily requests at 99.99% uptime.
- Developed interactive analytics dashboards processing 100M+ data points monthly.
- Collaborated with an 8-person frontend team, accelerating feature delivery by 25%.

SKILLS & TECHNOLOGIES

Languages JavaScript / TypeScript, Python

Backend	Node.js, Express.js, NestJS, FastAPI, CQRS, Domain-Driven Design, Microservices Architecture, RESTful APIs, GraphQL, gRPC, Protocol Buffers, WebRTC, WebSockets, MQTT, AMQP
Frontend	Next.js
Databases	PostgreSQL, MySQL, MongoDB, Redis, Kysely
DevOps / Cloud	Docker, AWS (S3, EC2, RDS, ECS, Lambda, SQS, Amplify), RabbitMQ, Mosquitto, Agile Methodologies

FEATURED PROJECTS

EduMarket

Proprietary

Education Services Marketplace • NestJS / GraphQL / PostgreSQL

- Bilingual (AR/EN) multi-vendor marketplace for nurseries, schools, tutors & transportation; CQRS + hexagonal modular monolith across 58 domain modules; drove a rebuild lifting throughput 6.8× and cutting p95 latency ~6× (79.5% → ~0% failures under load); event-driven via AWS SQS/RabbitMQ with Redis caching; Kashier/Paymob payments behind circuit breakers.

Snacky

github.com/MahmoudMetwalli/snacky

Full-Stack Food Ordering App • Next.js 14 / App Router

- High-performance food ordering app with real-time order tracking, admin dashboard, and mock payment integration (Stripe-ready).

Intelligent Product Catalog

github.com/MahmoudMetwalli/product_management

Smart Search Engine • Django / Python

- Advanced hybrid, typo-tolerant Arabic–English bilingual search using Elasticsearch, with full CRUD and admin dashboard for e-commerce backends.

PCI-DSS Payment Gateway

github.com/MahmoudMetwalli/payment-gateway

Enterprise Payment System • NestJS / TypeScript

- Secure card tokenization & 3D Secure, event-driven architecture with Outbox/Inbox patterns, guaranteed delivery via RabbitMQ, high-throughput transactions on MongoDB.

IoT Fleet Management

github.com/MahmoudMetwalli/gateway

Real-time IoT Platform • Express.js / Node.js

- Multi-tenant fleet platform managing thousands of gateways and millions of devices, with live telemetry via Socket.IO, OTA updates, and JWT auth.

EDUCATION & CERTIFICATIONS

AWS Solutions Architect Associate

Amazon Web Services

Scalable, highly available, fault-tolerant systems on AWS.

ALX Software Engineering Program

Holberton School | 2023–2024

Computer Science fundamentals and backend specialization.

McKinsey Forward Program

McKinsey & Company

Problem-solving, productivity, and digital leadership.

AWS Cloud Practitioner

Amazon Web Services

Foundational AWS Cloud concepts and security.

ALX Soft Skills Program

ALX Africa | 2023–2024

Professional development skills for the digital age.

Bachelor of Pharmacy

Ain Shams University, Cairo